# MTH101, Basic Linear Algebra

Monsoon 2025

**06/11/2025, 2:00PM-3:00PM.**

Tutorial-7

(1) Let $A$ be an $n \times n$ matrix.

   (i) Let $\alpha \in \mathbb{R}$, $X, Y \in \mathbb{R}^n$, $X_1 := \alpha X$ (scalar multiplication) and $Z := X + Y$. Verify that $AX_1 = \alpha AX$ and $AZ = AX + AY$.

   (ii) Verify that the set $\{X : \ AX = 0 \ \text{ for } \ X \in \mathbb{R}^n\}$ is a subspace of $\mathbb{R}^n$

   (iii) Verify that the set $\{AX \ : \ X \in \mathbb{R}^n\}$ is a subspace of $\mathbb{R}^n$.

(2) Let $A$ be an invertible $n \times n$ matrix. Prove that if $v_1, \ldots, v_m \in \mathbb{R}^n$ are $\mathbb{F}-$linearly independent vectors then $Av_1, \cdots, Av_n$ are also $\mathbb{F}-$linearly independent vectors.

(3) Let $\{v_1, \cdots, v_n\}$ be a set of $\mathbb{F}-$linearly independent vectors of a vector space $V$. Prove that if $v \in V$ and $v \notin \mathrm{Span}\{v_1, \cdots, v_n\}$ then $\{v, v_1, \cdots, v_n\}$ is an $\mathbb{F}-$linearly independent set.